

Sheet 9

A1.1

$$R(t + \Delta t) = R(t) + \Delta t \cdot \dot{R}(t) + \mathcal{O}(\Delta t^2)$$

$$f(x)_a = f(a) + f'(a)(x-a) + f''(a)\frac{(x-a)^2}{2} + \dots$$

$$\begin{aligned} \Rightarrow R(t + \Delta t)_t &= R(t) + \dot{R}(t)(t + \Delta t - t) + \frac{\ddot{R}(t)}{2} \Delta t^2 \dots \\ &= R(t) + \dot{R}(t) \Delta t + \mathcal{O}(\Delta t^2) \end{aligned}$$

$$\Rightarrow \frac{d}{dt} R(t + \Delta t) = \dot{R}(t) + \ddot{R}(t) \Delta t + \mathcal{O}(\Delta t^2)$$

A1.2

$$R(t - \Delta t) = R(t) - \Delta t \cdot \dot{R}(t)$$

$$\dot{R}(t - \Delta t) = \dot{R}(t) - \Delta t \cdot \ddot{R}(t)$$

$$R(t + \Delta t) = R(t) + \Delta t \cdot \dot{R}(t)$$

$$\rightarrow \text{one step back: } R(t + \Delta t - \Delta t) = R(t - \Delta t) + \Delta t \cdot \dot{R}(t - \Delta t)$$

$$= R(t) - \Delta t \cdot \dot{R}(t) + \Delta t \cdot \dot{R}(t) - \mathcal{O}(\Delta t^2)$$

$$= R(t) \quad ?$$